

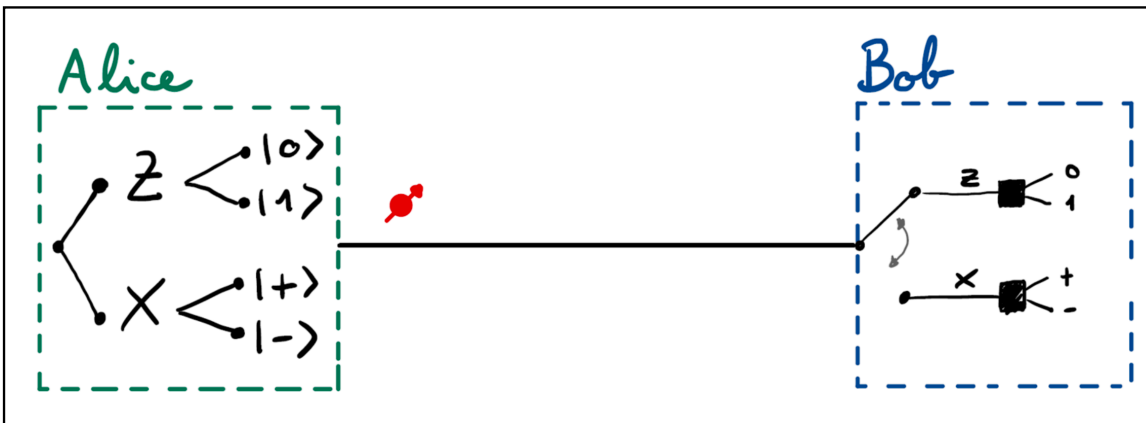
Bob Receives and Measures the State

- **Transmission:**
 - Alice sends $|\psi_A\rangle$ over the quantum channel to Bob.
 - Eve may interact with qubits but cannot know the basis b .
- **Bob receives the quantum state:**

$$\mathcal{E}(|\psi\rangle_A \langle\psi|), \quad \text{where} \quad |\psi\rangle_A = \bigotimes_{i=1}^{4n} |\psi_{a_i b_i}\rangle_A$$

- **Bob measures the state to get bit strings:**

$$a' = (a'_1, a'_2, \dots, a'_{4n}) \quad b' = (b'_1, b'_2, \dots, b'_{4n})$$



Alice

		Z		X		
		$ 0\rangle$	$ 1\rangle$	$ +\rangle$	$ -\rangle$	
Bob	Z	$ 0\rangle$	1	0	$\frac{1}{2}$	$\frac{1}{2}$
	$ 1\rangle$	0	1	$\frac{1}{2}$	$\frac{1}{2}$	
	X	$ +\rangle$	$\frac{1}{2}$	$\frac{1}{2}$	1	0
	$ -\rangle$	$\frac{1}{2}$	$\frac{1}{2}$	0	1	



BB84 Protocol: Sifting Step

- **Sifting Step:**

- Alice announces her basis string b .
- Bob announces the positions i where $b_i \neq b'_i$.
- Alice and Bob discard the bits $\{a_i, a'_i\}$ where $b_i \neq b'_i$.

Alice

- Alice's raw key: string a of length $2n$
- Bob's raw key: string a' of length $2n$
- After sifting: Approximately $2n$ bits remain

Bob

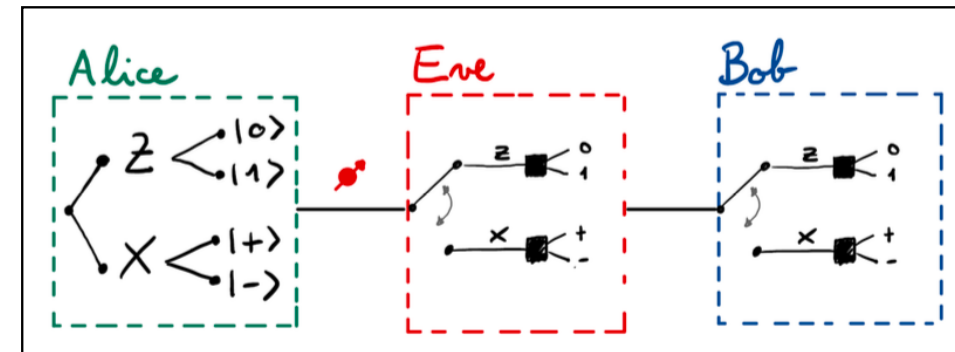
BB84 Protocol: Classical Post-Processing



- **Parameter Estimation:**
 - Use half of the sifted key to estimate the quantum bit error rate (QBER).
 - Detects potential eavesdropping by comparing sample bits.
- **Key Extraction:**
 - Apply **error correction** to reconcile Alice and Bob's remaining bits.
 - Apply **privacy amplification** to reduce Eve's potential knowledge.
 - Final shared key: m bits, where $m < n$

Intercept and Resend Attack

- Measurements affect the quantum state – can detect amount of eavesdropping and **abort if high**
- Copying unknown qubits is impossible (No-Cloning Thm)
- Intercept, Measure & Resend Attack:
- For the raw key: Alice and Bob used same basis ($x_i = y_i$)
- Eve measures in basis z_i
- With probability $p_1 = 1/2$ the basis $x_i \neq z_i$ (otherwise no eavesdropping is detected)
- After the measurement, Eve sends the output which is a state from the basis z_i
- Bob measures in the $x_i \neq z_i$ basis
- With prob. $p_2 = 1/2 = |\langle + | 0 \rangle|^2$ Bob obtains each of the two outcomes b_i i.e., with $p_2 = 1/2$ Bob obtains the different outcome from what Alice sent.
- Alice and Bob detect 25% QBER, i.e. $p_1 \times p_2 = 1/4$.



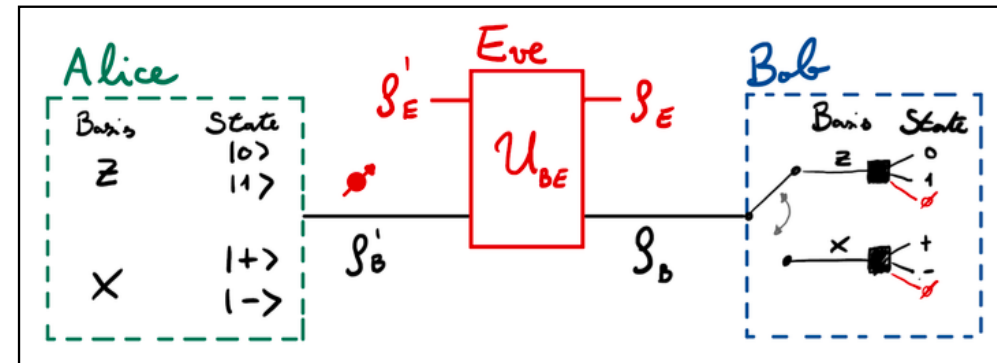
Alice

ZX

		$ 0\rangle$	$ 1\rangle$	$ +\rangle$	$ -\rangle$	
Bob	Z	$ 0\rangle$	$3/4$	$1/4$	$1/2$	$1/2$
	X	$ 1\rangle$	$1/4$	$3/4$	$1/2$	$1/2$
	Z	$ +\rangle$	$1/2$	$1/2$	$3/4$	$1/4$
	X	$ -\rangle$	$1/2$	$1/2$	$1/4$	$3/4$

Assumptions in QKD

- Quantum theory is correct and complete
- Authentic classical communication is possible
- Isolation of Alice's and Bob's labs
- The state prepared and measurement are characterised



- Types of attacks
 - Individual attack: Eve's attack is i.i.d. on all signal. They wait for the postprocessing phase to measure but the measurement is done on each ancilla independently
 - Collective attack: Eve's attack is still i.i.d. but all ancillas can be measured collectively.
 - Coherent attack: Eve can attack all states at the same time is available. The ancilla can be measured collectively.



Security Proof

- Full security proof $\Rightarrow$ all possible attacks of Eve.
- Alice: bit-string A ; Bob: bit-string B
- Eve: bit-string E the best guess she can make
- If QBER low then A, B more correlated than A, E or B, E : $I(A : B) > I(A : E)$
- Alice/Bob advantage in the final post-processing
- **Information Reconciliation (IR)**: Exchange information (error-correcting codes) to make $A=B$ (extra info leaked to Eve)
- **Privacy Amplification (PA)**: Distil shorter key completely secret from Eve (use universal hash functions to amplify privacy)



Realistic QKD

- Realistic systems have noise: $\text{QBER} \neq 0$ even if honest
- Cannot tell errors from noise Vs errors from eavesdropping
- QBER is used for:
 - Estimate correlation of Alice's raw bit-string A with Bob's B
 - Bound the max correlation that any adversary's bit string E can have with A
- If (A, B) "correlation" is higher than (A, E) then it is possible for Alice and Bob to distil an (identical) bit-string A totally secret from Eve (using IR & PA)
- The key-rate R , highest possible noise-tolerance and maximum distance possible all depend on the advantage $I(A : B) - I(A : E)$



SKR of DV-QKD Protocol

- SKR: $R = I(A : B) - I(A : E)$
- Let the QBER be Q .
- Then $I(A : B) = 1 - h_2(Q)$ (Max mutual information of Binary symmetric channel)
- Max information leaked to Eve: $I(A : E) \leq h_2(Q)$
- $R = 1 - h_2(Q) - h_2(Q) = 1 - 2h_2(Q)$
- Adding sifting and Efficiency factors:
 - Only half of the bits are kept after sifting $q = 1/2$
 - Error Correction is not perfect: Correction efficient factor $f \geq 1$
- **Realistic SKR:** $R = q[1 - fh_2(Q) - h_2(Q)]$
- QBER Threshold for positive SKR: $1 - 2h_2(Q) = 0$ gives $Q \approx 0.11$